HR ANALYTICS – PREDICT EMPLOYEE ATTRITION

Project Workflow Explained

1. **Data Loading**
   - The dataset (HREmployee_data.csv) is loaded using pandas.
   - A quick view of the data is shown using .head() to understand its structure.
2. **Initial Data Exploration**
   - You use .info(), .describe(), and .isnull().sum() to inspect data types, summary statistics, and check for missing values.
   - You explore specific columns like Attrition and Department to understand distribution and categories.
3. **Data Preprocessing**
   - Categorical features like Attrition are converted into numerical format using **Label Encoding**.
     - For example, 'Yes' becomes 1 and 'No' becomes 0.
   - You remove irrelevant or constant columns like EmployeeID, Over18, EmployeeCount, and StandardHours to avoid noise in the model.
4. **(Likely Next Steps)** – Though not yet visible in the preview:
   - **Feature Engineering** – Creating or modifying features to better capture patterns.
   - **Model Training** – Building a machine learning model (e.g., Decision Tree, Random Forest, or Logistic Regression).
   - **Model Evaluation** – Using accuracy, confusion matrix, ROC curve, etc., to assess performance.
   - **Interpretability** – Possibly SHAP or feature importance to understand what factors contribute to attrition.

Steps in the Project

1. **Data Loading & Exploration**
   - You load the HR dataset (HREmployee_data.csv) using pandas.
   - Initial checks (info, describe, isnull) help understand the data structure, types, and missing values.
   - You explore distributions like how many employees left, department-wise attrition, etc.

2. **Data Preprocessing**
    - Target variable Attrition is label encoded (Yes → 1, No → 0).
    - Irrelevant or constant columns like EmployeeID, Over18, etc., are dropped.
    - Categorical variables are converted to numeric using get_dummies (one-hot encoding).
3. **Data Visualization**
    - You use seaborn and matplotlib to analyze:
        - Attrition by department
        - Attrition vs. Salary Bands (based on DailyRate)
        - Attrition vs. promotions
        - Attrition vs. overtime
4. **Feature Engineering**
    - You create a new SalaryBand feature using quantile-based binning of DailyRate.
5. **Model Building**
    - You train a **Logistic Regression** model to predict attrition.
    - The dataset is split into training and test sets.
    - Features are scaled using StandardScaler for better model performance.
6. **Model Evaluation**
    - You evaluate the model using **accuracy score**, **confusion matrix**, and **classification report** to understand performance.

Attrition Prevention Suggestions

1. Monitor and Manage Workload (OverTime)

- **Insight**: High attrition is often associated with frequent overtime.
- **Action**:
    - Introduce workload balancing strategies.
    - Promote flexible working hours or hybrid work policies.
    - Offer compensatory time-off for employees working overtime.

2. Boost Career Growth Opportunities (Years Since Last Promotion)

- **Insight**: Employees with no recent promotions are more likely to leave.
- **Action**:

    - Implement clear career paths and regular promotion assessments.
    - Encourage internal job rotations and upskilling programs.
    - Recognize performance with both monetary and non-monetary rewards.

3. Improve Compensation Structure (Daily Rate / Salary Band)

- **Insight**: Lower salary bands show higher attrition rates.
- **Action**:
    - Benchmark salaries with industry standards.
    - Offer performance-based bonuses or benefits (e.g., wellness programs, stock options).

4. Engage Employees in Meaningful Work (Job Role & Department Insights)

- **Insight**: Certain departments or roles may experience higher attrition.
- **Action**:
    - Conduct regular satisfaction surveys in high-risk departments.
    - Encourage participation in strategic decision-making or innovation committees.

5. Enhance Employee Engagement

- **Insight**: Disengaged employees are more likely to leave even if other metrics look good.
- **Action**:
    - Offer mentoring programs, feedback sessions, and team-building activities.
    - Implement pulse surveys and act on feedback quickly.

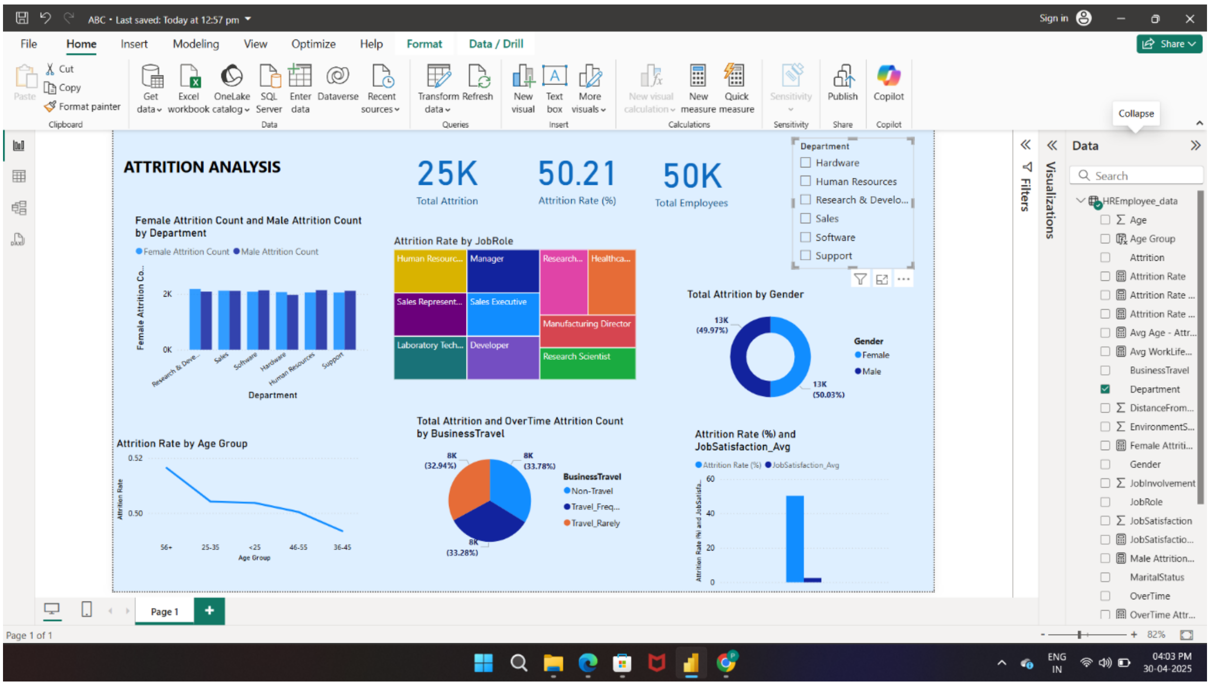
6. Support Work-Life Balance

- **Insight**: Overworked or stressed employees are prone to burnout.
- **Action**:
    - Provide access to counseling, mental health days, and wellness initiatives.
    - Train managers to detect signs of burnout and intervene early.

7. Exit Interview Analysis

- **Insight**: Valuable reasons for attrition can come from those who already left.
- **Action**:
    - Systematically analyze exit interview data for patterns.
    - Use this feedback to adjust retention strategies.

POWER BI DASHBOARD



Summary Metrics (KPIs)

- **Total Attrition:** 25K
  Total number of employees who left the company.
- **Attrition Rate (%):** 50.21%
  Indicates that over half of the workforce experienced attrition — a high figure, suggesting concern.
- **Total Employees:** 50K
  Total employee count (used to calculate attrition rate).

Key Visualizations

1. Female vs Male Attrition Count by Department (Bar Chart)

- Compares male and female attrition counts department-wise.
- Departments like **Research & Development** and **Sales** show high attrition in both genders.
- Gender-parity in attrition exists, but department-wise trends differ slightly.

2. Attrition Rate by Job Role (Treemap)

- Displays attrition rate by job roles.
- Larger, darker blocks indicate higher attrition roles:

- Roles like **Sales Representative** and **Human Resources** may have higher attrition.
- Roles like **Research Scientist** and **Developer** may show relatively lower attrition.

3. Total Attrition by Gender (Donut Chart)

- Very balanced attrition between **Female (49.97%)** and **Male (50.03%)**.
- Suggests gender is not a major factor in predicting attrition, based on raw numbers.

4. Attrition Rate by Age Group (Line Chart)

- Shows a decreasing trend in attrition as age increases:
  - Employees under 35 show the highest attrition.
  - Older groups (46+) show more stability — possibly due to job security or nearing retirement.

5. Attrition and OverTime Count by Business Travel (Pie Chart)

- Shows how attrition is distributed among business travelers:
  - **Travel_Frequently (33.78%)** and **Travel_Rarely (33.28%)** show similar levels of attrition.
  - Non-travelers also face similar rates — implies that travel frequency may not be a dominant attrition driver alone, but may combine with stress, overtime, or dissatisfaction.

6. Attrition Rate (%) vs Job Satisfaction (Bar Chart)

- Compares attrition with average job satisfaction scores.
- Lower job satisfaction is directly linked to higher attrition rates.
- Strong signal: improving job satisfaction could reduce attrition substantially.

Filters Panel (Right Side)

You have filters like:

- Department, BusinessTravel, Gender, etc., allowing you to drill down for specific insights.
- This helps in creating department-specific or demographic-specific intervention strategies.

Insights Summary

- **Age and job satisfaction** are clear risk indicators.
- **Departments like Sales and R&D** have high attrition, requiring targeted retention strategies.
- **Business travel frequency** does not seem to reduce attrition risk on its own.

- **Job role-level attrition** shows that some roles are more prone to churn — useful for HR planning.